DOCKER NETWORKING

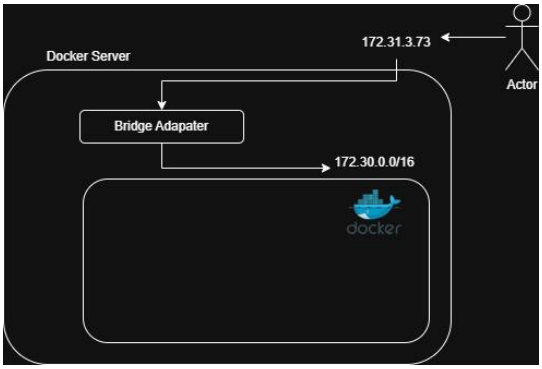
WHAT IS DOCKER NETWORK

Docker networking is a way to connect containers to each other and to the host system so they can communicate easily.

TYPES OF DOCKER NETWORKS

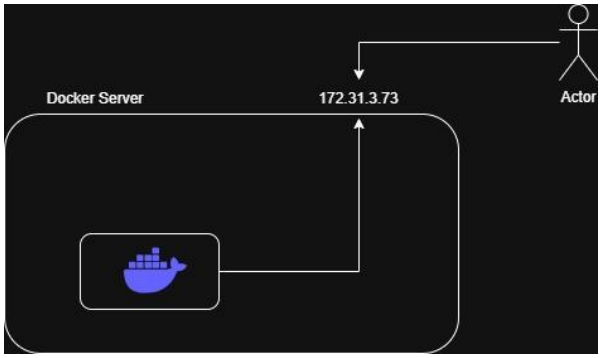
BRIDGE

1. Bridge Network is Docker's default private network.
2. It allows containers to communicate with each other internally on the same host.
3. Containers can also access the internet through the host machine.
4. External users can access containers using port mapping
5. **Example:**
   1.docker network create my_bridge
   2.docker network ls
   3. docker run -d --name containet1 --network my_bridge nginx



HOST

1. container directly uses the host machine's network.
2. Container shares the same IP address and ports as the host system.
3. Provides faster network performance but offers less security and isolation.
4. **Example:**
   docker run -d --name container1 --network host nginx



OVERLAY

1. Overlay network is used to connect containers running on different Docker hosts.
2. It creates a virtual private network between multiple Docker servers.
3. Containers can communicate securely even if they are on different machines or VPCs.
4. Mostly used in Docker Swarm and distributed applications.
5. Overlay network works like a tunnel between Docker hosts.

CONCEPT:

Suppose:

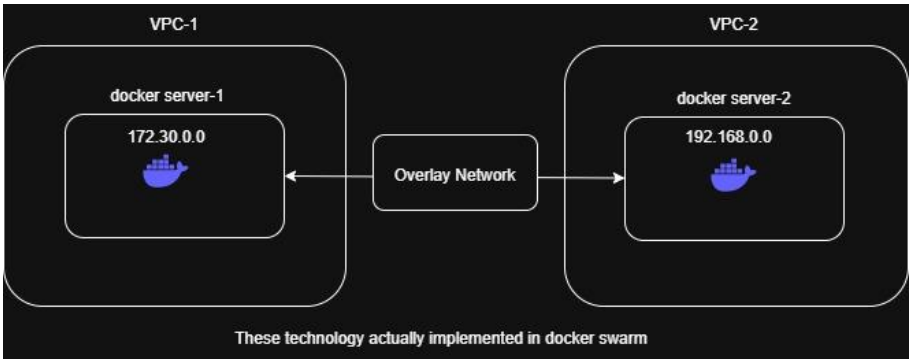
- Frontend container runs on Server-1
- Backend container runs on Server-2

Using overlay network:

- Both containers communicate like they are in the same local network.

COMMANDS

1. Initialize swarm
   docker swarm init
2. Create overlay network
   docker network create -d overlay my_overlay
3. Create service using overlay network
   docker service create --name web --network my_overlay nginx



IPVLAN

1. IPvlan allows containers to get separate IP addresses.
2. Containers share the host machine's MAC address.
3. It provides better performance and simpler network management.
4. Used in large-scale container environments.

MACVLAN

1. Macvlan allows containers to appear as physical devices on the network.
2. Each container gets its own MAC address and IP address.
3. Containers communicate directly with the physical network.
4. Used when containers need to behave like real servers.

NONE

1. No networking is provided to the container.
2. Container is fully isolated (no internet access and no communication with other containers).
3. Used for high security or testing applications without network access.
4. Commonly used for sensitive applications or databases that should not be exposed to the network.
5. There is no public as well as private ip to the container.
6. **Example:**
   docker run -d --name nginx_container --network none nginx



COMMANDS

1. **To check all the networks**
   docker network ls
2. **Delete Specific Network**
   docket network rm my_bridge
3. **Disconnect Container from Network**
   docker network disconnect my_bridge container1
4. **Delete Unused Networks**
   docker network prune
